Chemical equilibrium

Subject: Chemistry | Level: Class 11 | Curriculum: Sindh Board
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1. Topic Overview and Learning Objectives

Chemical equilibrium is one of the most significant chapters in your Class 11 Chemistry curriculum. It describes the state where the rate of the forward reaction equals the rate of the reverse reaction, leading to a dynamic balance in a closed system. Understanding this concept is essential because it governs how chemical reactions behave in real-world industrial processes, such as the production of ammonia or sulfuric acid.

In the context of the Sindh Board examinations, this topic carries substantial weight. Examiners frequently test your ability to derive the equilibrium constant expression, apply Le Chatelier's Principle to predict shifts in equilibrium, and solve numerical problems involving Kc and Kp. Mastering these concepts will not only help you score well in your theory papers but also provide a strong foundation for your future studies in physical chemistry.

The fundamental intuition behind equilibrium is that nature prefers stability. When a reaction reaches equilibrium, the concentrations of reactants and products remain constant over time, even though the reaction is still occurring at the molecular level. By learning to manipulate these conditions, chemists can optimize yields, which is why this topic is so crucial for chemical engineering and industrial chemistry applications.

2. Core Concepts and Detailed Revision Notes

2.1 Reversible and Irreversible Reactions

A chemical reaction is classified as either reversible or irreversible based on the directionality of the process. An irreversible reaction proceeds in only one direction until one of the reactants is completely consumed, often because the products escape the system as gases or form precipitates. In contrast, a reversible reaction can proceed in both the forward and reverse directions simultaneously. When a reversible reaction occurs in a closed container, it eventually reaches a state of chemical equilibrium. At this point, the rate at which reactants form products is exactly equal to the rate at which products reform reactants. It is vital to understand that equilibrium is dynamic, not static, meaning the reaction never truly stops; rather, the net concentrations of all species remain unchanged over time, provided the temperature and pressure are kept constant. This dynamic nature is a core concept for your board exams.

Key Points to Remember:

1. Irreversible reactions go to completion.
2. Reversible reactions never reach completion.
3. Equilibrium only occurs in closed systems.
4. Dynamic equilibrium implies equal rates of forward and reverse reactions.

2.2 Law of Mass Action and Kc

The Law of Mass Action states that the rate of a chemical reaction is directly proportional to the product of the active masses of the reactants, each raised to the power of its stoichiometric coefficient in the balanced chemical equation. This law allows us to derive the equilibrium constant expression, denoted as Kc. For a general reaction aA + bB yields cC + dD, the equilibrium constant is defined as the ratio of the product of the concentrations of the products to the product of the concentrations of the reactants, with each concentration raised to the power of its respective coefficient. The value of Kc is constant at a specific temperature. If Kc is very large, the equilibrium lies to the right, favoring products, whereas a very small Kc indicates that the reaction mixture contains mostly reactants at equilibrium.

Key Points to Remember:

1. Kc is temperature dependent.
2. Active mass is expressed in moles per cubic decimeter.
3. Pure solids and liquids are excluded from the Kc expression.
4. Kc provides information about the extent of a reaction.

2.3 Le Chatelier's Principle

Le Chatelier's Principle is a fundamental rule used to predict how a system at equilibrium will respond to external changes. It states that if a stress is applied to a system at equilibrium, the system will shift its position to counteract the effect of that stress. Common stresses include changes in concentration, pressure, volume, or temperature. For example, if you increase the concentration of a reactant, the system will shift to the right to consume the excess reactant and produce more product. Similarly, changing the pressure affects reactions involving gases; increasing pressure shifts the equilibrium toward the side with fewer moles of gas. Temperature changes are unique because they actually alter the value of the equilibrium constant itself. Understanding these shifts is critical for maximizing industrial production efficiency in chemical manufacturing.

Key Points to Remember:

1. Increasing reactant concentration shifts equilibrium to the right.
2. Increasing pressure favors the side with fewer gas moles.
3. Temperature change is the only factor that changes the Kc value.

4. Catalysts do not shift equilibrium; they only speed up the attainment.

2.4 Relationship between Kc and Kp

When dealing with gaseous reactions, it is often more convenient to express equilibrium in terms of partial pressures rather than molar concentrations. The equilibrium constant for partial pressures is denoted as Kp. The relationship between Kc and Kp is given by the equation Kp = Kc(RT)^delta n, where R is the ideal gas constant, T is the absolute temperature in Kelvin, and delta n is the change in the number of moles of gaseous products minus the number of moles of gaseous reactants. This derivation is essential for students because it allows us to convert between concentration-based equilibrium and pressure-based equilibrium. If delta n is zero, then Kp equals Kc. This mathematical relationship is frequently tested in board exams through numerical problems requiring the calculation of one constant given the other.

Key Points to Remember:

1. Kp is used for gaseous systems.
2. R is the universal gas constant (0.0821 L atm / mol K).
3. Delta n = moles of gaseous products - moles of gaseous reactants.
4. Kp = Kc when delta n = 0.

3. Key Definitions, Formulas, and Rules

1. **Law of Mass Action:** Rate is proportional to the product of active masses of reactants.
2. **Equilibrium Constant (Kc):** Kc = ([C]^c [D]^d) / ([A]^a [B]^b)
3. **Equilibrium Constant (Kp):** Kp = (Pc^c - Pd^d) / (Pa^a - Pb^b)
4. **Relationship Formula:** Kp = Kc(RT)^delta n
5. **Active Mass:** Concentration in mol/dm^3 represented by square brackets [ ].
6. **Dynamic Equilibrium:** State where forward and reverse reaction rates are equal.
7. **Le Chatelier Principle:** System shifts to counteract applied stress.
8. **Delta n:** Total moles of gaseous products minus total moles of gaseous reactants.

4. Step-by-Step Worked Examples

Example 1: Calculating Kc

Problem: For the reaction H2 + I2 yields 2HI, at equilibrium, the concentrations are [H2] = 0.1 M, [I2] = 0.2 M, and [HI] = 0.5 M. Calculate Kc.

Solution Steps:

Step 1: Write the expression: Kc = [HI]^2 / ([H2] - [I2])

Step 2: Substitute values: Kc = (0.5)^2 / (0.1 - 0.2)

Step 3: Calculate: Kc = 0.25 / 0.02

Step 4: Result: Kc = 12.5

Final Answer: Kc = 12.5

Example 2: Finding Kp from Kc

Problem: For N2 + 3H2 yields 2NH3, Kc is 0.5 at 500 K. Find Kp.

Solution Steps:

Step 1: Determine delta n: 2 - (1 + 3) = -2

Step 2: Use formula: Kp = Kc(RT)^delta n

Step 3: Substitute: Kp = 0.5 - (0.0821 - 500)^(-2)

Step 4: Calculate: Kp = 0.5 - (41.05)^(-2) = 0.5 / 1685.1

Final Answer: Kp = 0.000297

Example 3: Equilibrium Shift Prediction

Problem: In the reaction 2SO2 + O2 yields 2SO3 + heat, what happens if temperature is increased?

Solution Steps:

Step 1: Identify the reaction type: Exothermic

Step 2: Apply Le Chatelier: Increasing heat shifts equilibrium to absorb heat

Step 3: Direction: The reverse reaction absorbs heat

Step 4: Conclusion: Equilibrium shifts to the left

Final Answer: The equilibrium shifts to the left, decreasing the yield of SO3.

Example 4: Concentration Calculation

Problem: If Kc = 4 for A + B yields C + D, and [A]=1, [B]=1, [C]=2, find [D] at equilibrium.

Solution Steps:

Step 1: Write expression: 4 = ([C] - [D]) / ([A] - [B])

Step 2: Substitute: 4 = (2 - [D]) / (1 - 1)

Step 3: Solve for [D]: 4 = 2 - [D]

Step 4: [D] = 4 / 2

Final Answer: [D] = 2 M

5. Common Exam Mistakes to Avoid

1. Forgetting to raise concentrations to the power of their coefficients in the Kc expression.
2. Including pure solids or liquids in the equilibrium constant expression.
3. Confusing the direction of shift in Le Chatelier's Principle for exothermic versus endothermic reactions.
4. Failing to convert temperature to Kelvin when using the Kp = Kc(RT)^delta n formula.
5. Assuming that a catalyst changes the equilibrium constant value.
6. Miscalculating delta n by ignoring the coefficients of gaseous reactants or products.

6. Practice Questions and Model Answers

Question 1

Define chemical equilibrium and explain why it is called dynamic.

Answer 1: Chemical equilibrium is a state where the rates of forward and reverse reactions are equal. It is dynamic because the reactions continue to occur at the molecular level, but the net concentrations remain constant.

Question 2

How does a catalyst affect the position of equilibrium?

Answer 2: A catalyst does not affect the position of equilibrium. It increases the rate of both forward and reverse reactions equally, allowing the system to reach equilibrium faster.

Question 3

For the reaction PCl5 yields PCl3 + Cl2, how does increasing pressure affect the equilibrium?

Answer 3: Increasing pressure shifts the equilibrium to the side with fewer gas moles. Since the left side has 1 mole and the right side has 2 moles, the equilibrium shifts to the left.

Question 4

Write the Kc expression for 2NO2 yields N2O4.

Answer 4: Kc = [N2O4] / [NO2]^2

Question 5

What is the value of delta n for the reaction H2 + Cl2 yields 2HCl?

Answer 5: Delta n = 2 - (1 + 1) = 0.